

DOCUMENTATION ON INFERENTIAL STATISTICS

Inferential statistics is a branch of statistics that allows us to make predictions or inferences about a population based on a sample of data drawn from that population. While descriptive statistics summarize the data, inferential statistics helps us make generalizations and decisions under uncertainty. The main goal is to draw conclusions that extend beyond the immediate data alone.

Key Concepts in Inferential Statistics

1. **Population and Sample:**
 - **Population:** The complete set of items that interest an investigator.
 - **Sample:** A subset of the population selected for study.
2. **Parameter and Statistic:**
 - **Parameter:** A measurable characteristic of a population (e.g., population mean, population standard deviation).
 - **Statistic:** A measurable characteristic of a sample (e.g., sample mean, sample standard deviation).

Normal Distribution:

Normal distribution, also known as the Gaussian distribution, is a probability distribution that is symmetric about the mean, showing that data near the mean are more frequent in occurrence than data far from the mean.

Properties of Normal Distribution

1. Empirical Rule
2. Distortion in Normal Distribution
3. Central Limit Theorem
4. Standard Normal Distribution
5. Outliers
6. QQ plot
7. Log,Sqrt,Boxcox transformation

Empirical Rule:

- The empirical rule states that for a normal distribution, nearly all of the data will fall within three standard deviations of the mean. The empirical rule can be broken down into three parts:
 - 68% of data falls within the first standard deviation from the mean. (1 Sigma)
 - 95% fall within two standard deviations. (2 Sigma)
 - 99.7% fall within three standard deviations. (3 Sigma)
- Any points lying after 3 sigma is outliers.

Distortion in Normal Distribution:

The distortion in normally distributed curves can be quantified in 2 ways

1. Skewness
2. Kurtosis

Skewness in Normal Distribution

Skewness is asymmetry in a statistical distribution, in which the curve appears distorted or skewed either to the left or to the right. Skewness can be quantified to define the extent to which a distribution differs from a normal distribution

Kurtosis in Normal Distribution

In probability theory and statistics, kurtosis is a measure of the “peakedness” of the probability distribution of a real-valued random variable.

How much Skewness & Kurtosis

- If the skewness is between -0.5 and 0.5, the data are fairly symmetrical. If the skewness is between -1 and -0.5 or between 0.5 and 1, the data is moderately skewed.
- If the skewness is greater than 1 or less than -1, the data is highly skewed.
- A standard normal distribution has kurtosis of 3 and is recognized as mesokurtic. An increased kurtosis (>3) can be visualized as a thin “bell” with a high peak whereas a decreased kurtosis corresponds to a broadening of the peak and “thickening” of the tails.

Outliers:

An outlier is an observation point that is distant from other observations. An outlier may be due to variability in the measurement or it may indicate experimental error; the latter are sometimes excluded from the data set.

Central Limit Theorem:

The central limit theorem states that the distribution of sample means approximates a normal distribution as the sample size gets larger (assuming that all samples are identical in size), regardless of population distribution shape.

CLT in one sentence "Even if I'm not normal, the average is normal"

When collecting means of the samples from any distribution, the no of samples taken for calculating the mean should be greater or equal to 30.

Q-Q Plot

Q-Q plots are used to find the type of distribution for a random variable whether it be a Gaussian Distribution / Normal distribution or not.

Standard Normal Distribution:

The standard normal distribution is a special case of the normal distribution. It is the distribution that occurs when a normal random variable has a mean of zero and a standard deviation of one.

Z-Score / Z-Value / Standard Score

- A z-score (aka, a standard score) indicates how many standard deviations an element is above or below from the mean. A z-score can be calculated from the following formula.
- $z = (X - \mu) / \sigma$
- So on this normal curve, for $z = -1.22$, the area on the left is 0.1112 as we saw in the z-score table. Therefore, the probability that $x < 54$ is 11.12%.
- Z-score is the powerful tool to find the probability of distribution using Z-score table.

Outliers:

- Z-scores generally range from -3.0 to +3.0.
- - For bell shaped distributions, the empirical rule says 99.7% of all the data values have z-scores between -3.0 and +3.0.
- - We consider any z-score that is either less than -3.0 or greater than +3.0 to be an outlier.

Probability Density Function (PDF):

Probability density is the relationship between observations and their probability.

The overall shape of the probability density is referred to as a probability distribution, and the calculation of probabilities for specific outcomes of a random variable is performed by a probability density function, or PDF for short.

The probability density function for Normal distribution is given as

Cummulative Density Function (CDF): It provides a shortcut for calculating many probabilities at once. We integrate the pdf function to get the cumulative probability.

Covariance:

- It is the relationship between a pair of random variables where change in one variable causes change in another variable.
- It can take any value between $-\infty$ to $+\infty$, where the negative value represents the negative relationship whereas a positive value represents the positive relationship.

Correlation:

- It is the scaled version of Covariance.
- Correlation is a step ahead of covariance as it quantifies the relationship between two random variables. In simple terms, it is a unit measure of how these variables change concerning each other (normalized covariance value).

Hypothesis Testing:

Hypothesis:

- Hypothesis is a statement, assumption or claim about the value of the parameter (mean, variance, median etc.).
- A hypothesis is an educated guess about something in the world around you. It should be testable, either by experiment or observation.

Ex:-if we make a statement that "Dhoni is the best Indian Captain ever." This is an assumption that we are making based on the average wins and loses team had under his captaincy. We can test this statement based on all the match data.

Hypothesis Testing:

Comparing and analysing the relationships

- Does the treatment with new drug help more patients than the standard treatment with old drug?
- Which of these four methods is the most efficient way of teaching machine learning?

Types of Hypothesis Testing:

- When a hypothesis specifies an exact value of parameter, it is simple hypothesis. For eg., Motor cycle company claiming that a certain model gives an average mileage of 100km per litre, this is a case of simple hypothesis.
- If a hypothesis specifies a range of values then it is called a composite hypothesis. For eg., Average age of students in a class is greater than 20. This statement is a composite hypothesis.

Null Hypothesis: The null hypothesis is the hypothesis to be tested for possible rejection under the assumption that it is true. The concept of the null is similar to innocent until proven guilty.

Alternate Hypothesis: The alternative hypothesis complements the Null hypothesis. It is opposite of the null hypothesis such that both Alternate and null hypothesis together cover all the possible values of the population parameter.

Hypothesis Testing – Case Discussion

- Consider a court of law; the null hypothesis is that the defendant is innocent
- We require evidence to reject the null hypothesis (convict)
- When we collect evidence and try to reject null hypothesis, there are 2 errors that could potentially occur: Type 1 and Type 2 errors.

Type I Error and Type II Error

Critical Region:

- The critical region is that region in the sample space in which if the calculated value lies then we reject the null hypothesis.
- The critical region lies in one tail or two tails on the probability distribution curve according to the alternative hypothesis.
- The value of critical region is denoted by α .
- It is known as level of significance. i.e what is passing criteria of test.

Three Cases of Critical Region Arise:

- If the alternate hypothesis gives the alternate in both directions (less than and greater than) of the value of the parameter specified in null hypothesis, it is called Two tailed test.
- Here according to H_1 , mean can be greater than or less than 100. This is an example of Two tailed test.
- e.g. if H_0 : mean = 100 H_1 : mean not equal to 100
- If the alternate hypothesis gives the alternate in only one direction (either less than or greater than) of the value of the parameter specified in null hypothesis, it is called One tailed test.
- Similarly, if H_0 : mean ≥ 100 then H_1 : mean < 100
- Here, mean is less than 100, it is called One tailed test.

P-Value:

- Since we know that passing value of a test can be 1% or 5%. But we must also know what are the test score and this test score is known as p- value.
- Technically p-value (probability value) is the smallest level of significance at which a null hypothesis can be rejected.
- If p-value is greater than alpha, we do not reject the null hypothesis.
- If p-value is smaller than alpha, we reject the null hypothesis.

Process of Hypothesis Testing:

1. Set up Null hypothesis and Alternate hypothesis.
2. Decide the level of significance.(1% or 5%)
3. Select the test as per requirement.
4. Calculate the p-value.
5. If p-value less than level of significance, reject the null hypothesis.
6. If p-value more than level of significance, accept the null hypothesis.

Types of Hypothesis Tests:

- **Parametric Tests:**-Those test which considers the shape of distribution of sample.
- **Non – Parametric Tests:**-Those test which do not considers the shape of distribution of sample.

Parametric Hypothesis Test:

1. Z test
2. T/Student's T test
3. Paired t Test
4. One Way ANOVA

Non-Parametric Hypothesis Test:

1. Chi Square Test
2. Mann-Whitney Test
3. Wilcoxon Signed-Rank Test
4. Kruskal-Wallis Test
5. Friedman's ANOVA

T-Test:

- A t-test is an analysis of two populations means through the use of statistical examination; a t-test with two samples is commonly used with small sample sizes, testing the difference between the samples when the variances of two normal distributions are not known.

X -> Mean of sample set

μ -> Mean of Population

S -> Standard deviation of sample

N -> Sample size

Paired T-Test:

The paired t-test is performed when the samples typically consist of matched pairs of similar units, or when there are cases of repeated measures.

For example, there may be instances of the same patients being tested repeatedly—before and after receiving a particular treatment. In such cases, each patient is being used as a control sample against themselves.

One-Way Anova:

The one-way analysis of variance (ANOVA) is used to determine whether there are any statistically significant differences between the means of three or more independent (unrelated) group.

Chi-Square Test:

Chi-square test is used for categorical features in a dataset. We calculate Chi-square between each feature and the target and select the desired number of features with best Chi-square scores. It determines if the association between two categorical variables of the sample would reflect their real association in the population. Chi- square score is given by

Observed frequency = No. of observations of class

Expected frequency = No. of expected observations of class if there was no relationship between the feature and the target.